

Spotify

August 19, 2026

```
[1]: import pandas as pd
import numpy as np
```

```
[2]: twentythree = pd.read_json('Streaming_History_Audio_2023.json')
twentyfour = pd.read_json('Streaming_History_Audio_2024.json')
twentyfive_one = pd.read_json('Streaming_History_Audio_2025_1.json')
twentyfive = pd.read_json('Streaming_History_Audio_2025.json')
twentysix = pd.read_json('Streaming_History_Audio_2026.json')
```

```
[3]: spotify= pd.concat([
    twentythree,
    twentyfour,
    twentyfive,
    twentyfive_one,
    twentysix],
    ignore_index=True)
```

```
[4]: spotify.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 39252 entries, 0 to 39251
```

```
Data columns (total 23 columns):
```

#	Column	Non-Null Count	Dtype
0	ts	39252 non-null	object
1	platform	39252 non-null	object
2	ms_played	39252 non-null	int64
3	conn_country	39252 non-null	object
4	ip_addr	39252 non-null	object
5	master_metadata_track_name	39096 non-null	object
6	master_metadata_album_artist_name	39096 non-null	object
7	master_metadata_album_album_name	39096 non-null	object
8	spotify_track_uri	39096 non-null	object
9	episode_name	41 non-null	object
10	episode_show_name	41 non-null	object
11	spotify_episode_uri	41 non-null	object
12	audiobook_title	115 non-null	object
13	audiobook_uri	115 non-null	object

```

14 audiobook_chapter_uri          115 non-null    object
15 audiobook_chapter_title        81 non-null    object
16 reason_start                  39252 non-null object
17 reason_end                    39252 non-null object
18 shuffle                      39252 non-null bool
19 skipped                      39252 non-null bool
20 offline                      39252 non-null bool
21 offline_timestamp              39204 non-null float64
22 incognito_mode                39252 non-null bool
dtypes: bool(4), float64(1), int64(1), object(17)
memory usage: 5.8+ MB

```

```

[5]: spotify=spotify.drop(['episode_name', 'episode_show_name',
    ↪ 'spotify_episode_uri', 'audiobook_title', 'audiobook_uri',
    ↪ 'audiobook_chapter_uri', 'audiobook_chapter_title', 'offline_timestamp'],
    ↪ axis=1)

```

```

[6]: spotify= spotify.rename(columns={'ts': 'timestamp',
    'conn_country': 'country',
    'ip_addr': 'ip_address',
    'master_metadata_track_name': 'track_name',
    'master_metadata_album_artist_name': 'artist_name',
    'master_metadata_album_album_name': 'album_name',
    'spotify_track_uri': 'spotify_uri',
    })

```

```

[7]: spotify=spotify.drop_duplicates()

```

```

[8]: spotify=spotify.dropna()

```

```

[9]: spotify['timestamp'] = pd.to_datetime(spotify['timestamp'], errors="coerce")

```

```

[10]: spotify['timestamp']=spotify['timestamp'].dt.tz_convert('America/Los_Angeles')

```

```

[11]: spotify['date'] = spotify['timestamp'].dt.date
spotify['year'] = spotify['timestamp'].dt.year
spotify['month'] = spotify['timestamp'].dt.month
spotify['day_of_week'] = spotify['timestamp'].dt.day_name()
spotify['hour'] = spotify['timestamp'].dt.hour

```

```

[12]: spotify

```

```

[12]:          timestamp platform  ms_played country \
0      2023-03-30 06:00:15-07:00      ios      2922      US
2      2023-08-18 22:00:32-07:00      ios      6592      US
5      2023-08-20 09:34:01-07:00      ios     176579      US

```

6	2023-08-20 09:34:42-07:00	ios	38425	US
7	2023-12-15 14:05:12-08:00	ios	7168	US
...	...	...	...	...
39247	2026-06-10 12:51:08-07:00	ios	196831	US
39248	2026-06-10 12:54:35-07:00	ios	203361	US
39249	2026-06-10 14:27:50-07:00	ios	249034	US
39250	2026-06-10 14:32:15-07:00	ios	179268	US
39251	2026-06-10 15:09:28-07:00	ios	76906	US

	ip_address	track_name	\
0	166.199.5.83	Still I Rise	
2	141.164.113.50	august	
5	141.164.113.50	Dance The Night - From Barbie The Album	
6	141.164.113.50	As It Was	
7	141.164.113.50	I Write Sins Not Tragedies	
...	...	...	
39247	2600:387:15:80e::14	Hold. Wait. Fall.	
39248	2600:387:15:80e::14	Losing Sleep	
39249	2600:387:15:80e::14	Hot Blooded	
39250	2600:387:15:80e::14	The Only One	
39251	2600:387:15:80e::14	zZz	

	artist_name	album_name	\
0	2Pac	Still I Rise	
2	Taylor Swift	folklore	
5	Dua Lipa	Dance The Night (From Barbie The Album)	
6	Harry Styles	Harry's House	
7	Panic! At The Disco	A Fever You Can't Sweat Out	
...	...	...	
39247	Bronze Whale	Hold. Wait. Fall.	
39248	WDL	No Wings Airline	
39249	New Constellations	It Comes in Waves	
39250	RITUAL	Every Night Another But Not You	
39251	DROELOE	zZz	

	spotify_uri	reason_start	\
0	spotify:track:44AiTNJgT1LTz3hAe0jng9	playbtn	
2	spotify:track:3hUxzQpSfdDqwM3ZTFQYOK	trackdone	
5	spotify:track:1vYXt7VSjH9JIM5oRRo7vA	trackdone	
6	spotify:track:4Dvkj6JhhA12EX05fT7y2e	trackdone	
7	spotify:track:4bPQsOPHn4xbipzdPfn6du	playbtn	
...	...	...	
39247	spotify:track:7txmM435zxM1UuGBxd8y4D	trackdone	
39248	spotify:track:2UmJxYuEEibRZ7WsVWvpW0	trackdone	
39249	spotify:track:1CLZzeLWqdhz5IfR5vHy4u	trackdone	
39250	spotify:track:3gZcnt0Vys6X9bSGHv9Mpk	trackdone	
39251	spotify:track:29Uip5vvwAeeNiT8knrmyC	trackdone	

	reason_end	shuffle	skipped	offline	\
0	logout	True	False	False	
2	endplay	False	True	False	
5	trackdone	False	False	False	
6	unexpected-exit-while-paused	False	False	False	
7	logout	True	False	False	
...	...	...	...	...	
39247	trackdone	False	False	False	
39248	trackdone	False	False	False	
39249	trackdone	False	False	False	
39250	trackdone	False	False	False	
39251	unexpected-exit-while-paused	False	False	False	

	incognito_mode	date	year	month	day_of_week	hour
0	False	2023-03-30	2023	3	Thursday	6
2	False	2023-08-18	2023	8	Friday	22
5	False	2023-08-20	2023	8	Sunday	9
6	False	2023-08-20	2023	8	Sunday	9
7	False	2023-12-15	2023	12	Friday	14
...	...	...	...	...	...	...
39247	False	2026-06-10	2026	6	Wednesday	12
39248	False	2026-06-10	2026	6	Wednesday	12
39249	False	2026-06-10	2026	6	Wednesday	14
39250	False	2026-06-10	2026	6	Wednesday	14
39251	False	2026-06-10	2026	6	Wednesday	15

[38977 rows x 20 columns]

```
[13]: artist_top_10 = spotify['artist_name'].value_counts().head(10)
      artist_top_10
```

```
[13]: artist_name
      ODESZA          3482
      Twenty One Pilots 1657
      BRONSON          901
      Big Wild         760
      Bob Moses        609
      Crooked Colours   570
      MEMBA            513
      Parry Gripp       510
      PINES            481
      Linkin Park       412
      Name: count, dtype: int64
```

```
[14]: album_top_10= spotify['album_name'].value_counts().head(10)
      album_top_10
```

```
[14]: album_name
      BRONSON                      858
      A Moment Apart              741
      The Last Goodbye            716
      Trench                      630
      In Return                   500
      Clancy                     442
      Summer's Gone              417
      A Moment Apart (Deluxe Edition) 341
      Blurryface                 306
      Superdream                 275
      Name: count, dtype: int64
```

```
[15]: track_top_10= spotify['track_name'].value_counts().head(10)
      track_top_10
```

```
[15]: track_name
      It's Only - ODESZA VIP Remix  146
      Better Now                   143
      A Moment Apart               135
      All My Life                  134
      Selfish Soul - ODESZA Remix  130
      BLINE                       128
      Heavier                     127
      Water                       122
      TENSE                       120
      Bloodfeather                119
      Name: count, dtype: int64
```

```
[20]: album_top_10_2023 = (
      spotify[spotify['year'] == 2023]['album_name'].value_counts().head(10)
      )
      album_top_10_2023
```

```
[20]: album_name
      With Teeth                   6
      The Fragile                 5
      The Downward Spiral         4
      A Fever You Can't Sweat Out 3
      Lateralus                   3
      Still                       2
      Lost Highway                2
      ††† (Crosses)               2
      Enima                      2
      Mer De Noms                 2
      Name: count, dtype: int64
```

```
[16]: artist_top_10_2023 = (
        spotify[spotify['year'] == 2023]['artist_name'].value_counts().head(10)
    )
    artist_top_10_2023
```

```
[16]: artist_name
Nine Inch Nails      19
T00L                  6
A Perfect Circle     5
Panic! At The Disco  4
Deftones             4
    (Crosses)         2
Nirvana              2
Jane's Addiction     2
Porcupine Tree       2
2Pac                 1
Name: count, dtype: int64
```

```
[21]: track_top_10_2023 = (
        spotify[spotify['year'] == 2023]['track_name'].value_counts().head(10)
    )
    track_top_10_2023
```

```
[21]: track_name
I Write Sins Not Tragedies      3
Ruiner                          3
All The Love In The World      2
The Perfect Drug                2
The Hand That Feeds            2
And All That Could Have Been    2
The Wretched                    2
As It Was                      1
Dance The Night - From Barbie The Album  1
Misery Business                 1
Name: count, dtype: int64
```

```
[22]: album_top_10_2024 = (
        spotify[spotify['year'] == 2024]['album_name'].value_counts().head(10)
    )
    album_top_10_2024
```

```
[22]: album_name
Trench      548
BRONSON     321
The Last Goodbye  320
Clancy       276
A Moment Apart  247
```

Blurryface	190
In Return	151
The Silence in Between	121
Summer's Gone	121
Mister Asylum	120

Name: count, dtype: int64

```
[23]: artist_top_10_2024 = (
        spotify[spotify['year'] == 2024]['artist_name'].value_counts().head(10)
    )
    artist_top_10_2024
```

```
[23]: artist_name
Twenty One Pilots    1045
ODESZA              1038
BRONSON             332
Bob Moses           284
Big Wild            253
MEMBA              236
Highly Suspect      228
pluko               207
Crooked Colours     204
Chevelle            195
Name: count, dtype: int64
```

```
[24]: track_top_10_2024 = (
        spotify[spotify['year'] == 2024]['track_name'].value_counts().head(10)
    )
    track_top_10_2024
```

```
[24]: track_name
TENSE                86
TIME + ENERGY       72
Bloodfeather         68
Stand Off            67
All My Life          66
Nico and the Niners  62
SUB ZERO             60
Age of Machine       58
Jumpsuit             57
HEART ATTACK         54
Name: count, dtype: int64
```

```
[25]: album_top_10_2025 = (
        spotify[spotify['year'] == 2025]['album_name'].value_counts().head(10)
    )
    album_top_10_2025
```

```
[25]: album_name
      BRONSON                      434
      A Moment Apart              422
      The Last Goodbye            346
      In Return                   306
      Summer's Gone               239
      A Moment Apart (Deluxe Edition) 182
      Superdream                  143
      Invincible EP               126
      Clancy                      120
      Vera                       119
      Name: count, dtype: int64
```

```
[18]: artist_top_10_2025 = (
      spotify[spotify['year'] == 2025]['artist_name'].value_counts().head(10)
      )
      artist_top_10_2025
```

```
[18]: artist_name
      ODESZA                      2072
      Parry Gripp                 493
      BRONSON                     462
      Big Wild                    456
      Twenty One Pilots           400
      PINES                       277
      Bob Moses                   268
      Crooked Colours             248
      MEMBA                       235
      BAYNK                       190
      Name: count, dtype: int64
```

```
[26]: track_top_10_2025 = (
      spotify[spotify['year'] == 2025]['track_name'].value_counts().head(10)
      )
      track_top_10_2025
```

```
[26]: track_name
      Heavier                      98
      In The Rain                  87
      Better Now                   85
      It's Only - ODESZA VIP Remix 85
      A Moment Apart              78
      CALL OUT                     77
      affection                    75
      Fate                        74
      Selfish Soul - ODESZA Remix  74
      American Money               72
```


Name: count, dtype: int64

```
[27]: album_top_10_2026 = (  
        spotify[spotify['year'] == 2026]['album_name'].value_counts().head(10)  
    )  
    album_top_10_2026
```

```
[27]: album_name  
BRONSON                103  
No Wings Airline      96  
Levi Don't Do It      85  
Breach                84  
FLUX                  76  
Runaway (U & I)       73  
A Moment Apart        72  
Langata               69  
Hear Me Out           67  
Inhale / Exhale       65  
Name: count, dtype: int64
```

```
[19]: artist_top_10_2026 = (  
        spotify[spotify['year'] == 2026]['artist_name'].value_counts().head(10)  
    )  
    artist_top_10_2026
```

```
[19]: artist_name  
ODESZA                 372  
Twenty One Pilots     212  
Phantogram            207  
Linkin Park           200  
Ian Asher             155  
RÜFÜS DU SOL          139  
PLÜM                  131  
Crooked Colours       118  
LEAP                  116  
BRONSON               107  
Name: count, dtype: int64
```

```
[28]: track_top_10_2026 = (  
        spotify[spotify['year'] == 2026]['track_name'].value_counts().head(10)  
    )  
    track_top_10_2026
```

```
[28]: track_name  
Lost My Mind (Wake Up)    83  
go steady                 76  
Runaway (U & I)           73
```

4am	62
I C Light	61
God Needs The Devil	61
Over & Over	61
Losing Sleep	60
In My Head	56
oh no i hope i don't fall	55
Name: count, dtype: int64	

[]: